

BRADFORD, PA, USA

WMO: 725266

Lat: 41.800N

Lon: 78.633W

Elev: 653

StdP: 93.72

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04751

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.0	-16.3	-23.3	0.5	-18.0	-20.6	0.6	-14.9	10.3	-9.8	9.2	-8.3	2.8	270	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	28.6	20.4	27.3	19.6	26.1	19.0	22.0	26.0	21.2	25.1	20.4	24.1	3.5	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	16.7	23.7	19.9	15.8	23.0	19.1	15.0	22.3	67.5	26.0	64.3	24.8	61.5	24.2	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.9	7.1	DB	-23.9	30.5	3.8	1.7	-26.6	31.7	-28.8	32.7	-30.9	33.7	-33.7	34.9	(4)
(5)			WB	-24.1	23.4	3.7	1.1	-26.8	24.2	-28.9	24.8	-31.0	25.4	-33.6	26.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.4	-5.4	-4.1	0.1	6.8	12.8	16.8	18.8	18.1	14.7	8.7	3.0	-2.2		
	DBStd	9.92	6.84	6.15	6.23	5.24	4.76	3.64	3.09	2.97	3.94	4.85	5.19	5.60		
	HDD10.0	2027	479	397	314	123	26	1	0	0	5	85	218	379		
	HDD18.3	4123	736	629	567	347	182	71	33	41	121	300	461	636		
	CDD10.0	1077	1	1	7	27	112	205	273	252	148	44	7	2		
	CDD18.3	129	0	0	0	1	8	24	47	34	13	2	0	0		
	CDH23.3	1134	0	0	1	13	110	224	409	258	111	8	0	0		
	CDH26.7	176	0	0	0	1	14	33	77	35	16	0	0	0		
(14) Wind	WSAvg	3.0	3.7	3.7	3.8	3.5	2.9	2.5	2.1	2.0	2.2	2.9	3.2	3.5		
(15) Precipitation	PrecAvg	1127	72	69	85	87	103	116	108	97	105	81	95	94		
	PrecMax	1500	148	134	159	158	224	285	256	176	196	173	227	185		
	PrecMin	674	10	18	13	33	24	21	35	42	39	18	10	31		
	PrecStd	184	35	34	39	33	48	62	44	41	43	32	43	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	14.8	21.3	25.3	28.4	29.2	30.5	29.4	28.7	24.4	18.6	15.2		
		MCWB	11.2	10.2	13.5	15.0	18.7	20.3	21.7	20.9	20.6	17.9	12.1	11.3		
	2%	DB	11.0	10.6	16.5	22.3	26.3	27.7	28.7	27.5	26.3	21.7	15.7	12.1		
		MCWB	9.4	7.2	10.4	13.1	17.1	19.9	21.0	20.0	19.3	15.6	10.8	10.1		
	5%	DB	7.5	7.7	13.1	19.2	24.1	26.1	27.2	26.2	24.3	19.5	13.6	8.7		
		MCWB	5.5	4.6	9.2	11.4	16.1	19.1	20.0	19.4	18.2	15.0	10.2	6.4		
	10%	DB	4.2	4.9	10.0	16.4	21.9	24.2	25.8	24.6	22.3	17.2	11.7	5.7		
		MCWB	2.4	2.4	6.1	9.8	14.9	17.8	19.2	18.6	17.2	13.7	8.9	3.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.9	11.6	14.2	16.6	20.3	21.9	23.3	22.8	22.4	19.7	14.4	13.0		
		MCDB	12.7	13.3	20.1	22.8	25.9	26.3	28.3	27.0	26.4	22.5	16.7	14.8		
	2%	WB	9.5	7.9	11.8	14.7	19.0	21.0	22.2	21.6	20.7	17.8	12.5	10.1		
		MCDB	10.9	10.0	14.8	19.4	23.9	25.6	26.7	25.0	24.5	20.2	14.3	11.6		
	5%	WB	6.0	5.1	9.5	13.0	17.7	20.2	21.3	20.7	19.6	15.9	10.7	6.6		
		MCDB	7.4	7.2	12.7	17.3	22.1	24.3	25.3	24.0	22.6	18.2	12.9	8.3		
	10%	WB	2.6	2.4	6.5	11.2	16.5	19.3	20.4	19.9	18.4	14.0	8.7	3.9		
		MCDB	4.2	4.8	9.8	15.5	20.5	22.8	23.9	23.2	21.0	16.6	11.4	5.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.8	9.7	11.9	12.5	11.8	12.1	11.6	11.8	10.2	8.3	6.9		
		MCDBR	9.6	11.8	14.0	16.7	15.7	13.9	13.4	13.2	13.9	13.9	11.7	10.5		
	5% WB	MCWBR	8.6	9.1	9.0	9.3	7.8	7.1	6.6	6.7	7.5	8.1	8.3	9.0		
		MCWBR	9.4	11.3	12.4	13.2	12.4	11.4	11.2	10.6	10.6	10.9	9.8	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.288	0.313	0.327	0.349	0.378	0.418	0.417	0.409	0.354	0.343	0.303	0.286		
		taud	2.406	2.325	2.398	2.396	2.330	2.238	2.247	2.263	2.409	2.447	2.528	2.475		
	Ebn at Noon		864	890	922	920	897	858	854	850	880	848	845	839		
		Edh at Noon	79	100	105	114	125	137	135	129	103	89	70	68		
(44) All-Sky Solar Radiation	RadAvg		1.41	2.19	3.25	4.38	5.12	5.56	5.67	5.08	4.09	2.60	1.68	1.11		
	RadStd		0.12	0.26	0.35	0.47	0.52	0.46	0.44	0.30	0.45	0.27	0.24	0.14		

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+111	+34	(46)
(47) Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+89	N/A	(47)

Nomenclature: See separate page